DVMDOSTEM v0.8.2: a terrestrial ecosystem model designed to represent arctic, boreal and permafrost ecosystem dynamics

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Summary

The impacts of climate change on natural ecosystems are the result of complex physical and ecological processes operating and interacting at a variety of spatio-temporal scales, that can be represented in process-based ecosystem models.

DVMDOSTEM is an advanced process-based terrestrial ecosystem model (TEM) designed to study ecosystem responses to climate changes and disturbances. It has a particular focus on permafrost regions (i.e. regions characterized by soils that stay partially frozen all year round for at least two consecutive years), encompassing boreal, arctic, and alpine landscapes. The model couples a dynamic vegetation module (DVM) and a dynamic organic soil (DOS) module to simulate processes at yearly and monthly scales, with some physical processes operating at an even finer temporal resolution. Its versatility allows for site-specific to regional simulations, making it valuable for predicting shifts in permafrost, vegetation, and carbon (C) and nitrogen (N) dynamics. While DVMDOSTEM has been described in the methods sections of many manuscripts, this paper is the first stand alone description of DVMDOSTEM, independent of a particular scientific investigation.

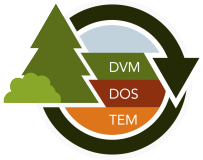

Figure 1: Logo for DVMDOSTEM

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Statement of need

Arctic and boreal regions underlain by permafrost store nearly half of the world's soil organic C - approximately 1,440-1,600 Pg (Hugelius et al., 2014; Edward A. G. Schuur et al., 2022). These regions are warming four times faster than the rest of the globe, driving widespread and rapid permafrost thaw (Rantanen et al., 2022; Smith et al., 2022). As permafrost thaws, soil organic C becomes available for decomposition and release as greenhouse gasses (GHGs) to the atmosphere. Climate-driven permafrost thaw and the associated release of GHGs can influence the global climate system, a phenomenon called the permafrost carbon-climate feedback or PCCF (Koven et al., 2011; E. A. G. Schuur et al., 2015). The PCCF has been identified as one of the largest sources of uncertainty in future climate projections and therefore needs to be accurately represented in global earth system models (Schädel et al., 2024). The `DVMDOSTEM` ecosystem model has been developed with particular attention to physical and biological processes in high latitude systems and is therefore well suited to assessing and informing our understanding of the PCCF.

Model Design

`DVMDOSTEM` is designed to simulate the key biophysical and biogeochemical processes between the soil, the vegetation and the atmosphere. The evolution and refinement of `DVMDOSTEM` have been shaped by extensive research programs and applications both in permafrost and non-permafrost regions (Briones et al., 2024; Eugénie S. Euskirchen et al., 2022; H. Genet et al., 2013; Hélène Genet et al., 2018; Jafarov et al., 2013; Yi et al., 2009, 2010). The model is spatially explicit and represents ecosystem response to climate and disturbances at seasonal (i.e. monthly) to centennial scales.

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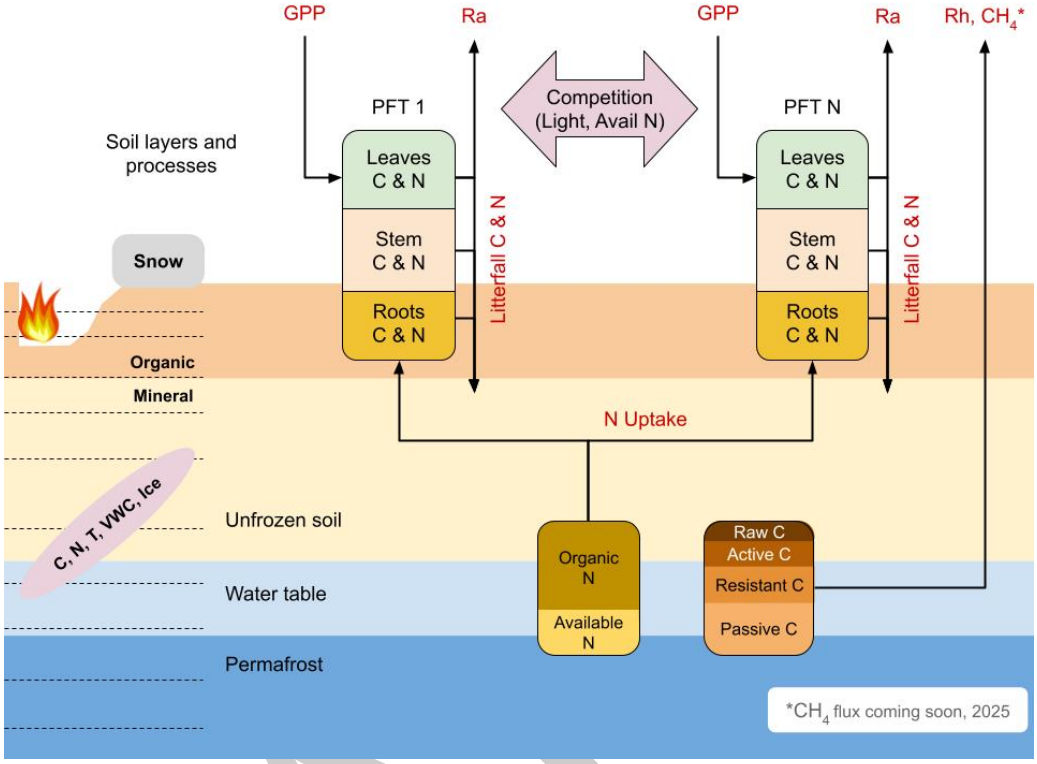

Figure 2: Overview of DVMDOSTEM soil and vegetation structure. On the left is the soil structure showing the layers and different properties that are tracked (purple bubble: carbon (C), nitrogen (N), temperature (T), volumetric water content (VWC), ice). Each of the layers with properties described above is also categorized as organic (fibric or humic) or mineral. Additionally, the model simulates snow layers and the removal of soil organic layers due to fire. On the right is the vegetation structure showing plant functional types (PFTs) within a community type (CMT) and the associated pools and fluxes of C and N. Each PFT is split into compartments (leaf, stem and root) which track their own C and N content and associated fluxes. The fluxes are represented with red text while the pools are black. In addition, there is competition among the PFTs for light, water, and available N, shown with the purple arrow in the top center.

The snow and soil columns are split into a dynamic number of layers to represent their impact on thermal and hydrological dynamics and the consequences for soil C and N dynamics. Vegetation composition is modeled using community types (CMTs), each of which consists of multiple plant functional types (PFTs - groups of species sharing similar ecological traits). This structure allows the model to represent the effect of competition for light, water and nutrients on vegetation composition (E. S. Euskirchen et al., 2009), as well as the role of nutrient limitation on permafrost ecosystem dynamics, with coupling between C and N cycles (E. S. Euskirchen et al., 2009; McGuire et al., 1992). Finally, the model represents the effects of wildfire in order to evaluate the role of climate-driven fire intensification on ecosystem structure and function (H. Genet et al., 2013; Yi et al., 2010). The structure of DVMDOSTEM is represented visually in Figure 2.

Acknowledgements

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